

What's happening to the Milky Way?

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Introduction (1)

The Milky Way's disc is not a static, flat plane; it is actively bending and rippling. Building on previous work [1], this project investigates the Galactic warp and its dynamics using an unprecedented dataset of ~ 8.7 million Red Clump stars from Gaia DR3 and WISE. For the first time, precessing disc models are applied to this highly precise standard-candle population.

To maximise spatial coverage, missing line-of-sight velocities for over 4 million stars are estimated by assuming zero Galactocentric radial velocity. By testing both a fixed and radially 'twisted' Line of Nodes (LON)¹, the aim is to understand the warp's influence on stars within the disc and uncover its geometry and structural evolution over time.

These results can ultimately be compared to theoretical models and existing literature to better understand the physical origin of the warp. By doing so, it becomes possible to better constrain leading theories of what excited the warp and allow for a more definitive distinction between long-lived structural features and transient, impulse-like perturbations.

Modelling (3)

The warp shape model assumes an antisymmetric, rigidly precessing disc where the contribution of radial motion to vertical velocity is negligible. The vertical position of stars depends on both R and ϕ such that:

$$z_0(R, \phi) = h(R) \sin(\phi - \phi'_w(R) - \omega_p(R)t).$$

Average vertical velocities observed from the data are compared to vertical velocities predicted from Jeans' first equation. The expected vertical velocity ($\bar{v}_{z,mod}$) is given by:

$$\bar{v}_{z,mod} = \left(\frac{\bar{v}_\phi}{R} - \omega_p(R) \right) h(R) \frac{\partial}{\partial \phi} (\sin(\phi - \phi'_w(R) - \omega_p(R)t))$$

Two different configurations are tested:

1. Fixed Line of Nodes: The warp orientation ($\phi'_w(R)$) and precession rate ($\omega_p(R)$) are constant across the entire disc.
2. Twisted Line of Nodes: Both warp orientation and precession rate are allowed to vary linearly with Galactocentric radius such that:

$$\phi'_w(R) = \phi_{w,0} + \phi_{w,1}(R - R_s); \quad \omega_p(R) = \omega_{p,0} + \omega_{p,1}(R - R_s)$$

To constrain the physical parameters of the Galactic warp, Markov Chain Monte Carlo (MCMC) analysis was used via the Python package emcee [5]. Walkers² explore the parameter space to fit the data to the models, providing robust posterior probability distributions and precise uncertainty measurements for the warp's amplitude (h), precession rate (ω_p), and line of nodes (ϕ'_w).

$h(R) \rightarrow$ Warp Amplitude : $\phi \rightarrow$ Galactocentric azimuth : $\phi_{w,0}, \omega_{p,0} \rightarrow$ Base LON and Precession Rate : $\phi'_w \rightarrow$ Line of Nodes : $\phi_{w,1}, \omega_{p,1} \rightarrow$ LON and Precession Rate Gradients : $R_s \rightarrow$ Reference radius fixed at 12 kpc.

Conclusion (5)

Both models reproduce vertical velocity trends, confirming a precessing, dynamical warp. The twisted model's height aligns better with literature, whereas the fixed model predicts a much steeper warp (Figure 6). The prograde precession rate $\sim 12 \text{ km s}^{-1} \text{ kpc}^{-1}$ matches giant stars [6,7] but slightly exceeds younger Cepheids [3]. Furthermore, the derived line of nodes ($\sim 140^\circ$) is offset from the $\sim 165^\circ$ found in recent literature [8]. It must be noted that while the global twisted model yields a trailing LON spiral (Figure 5), analysing discrete 2 kpc annuli from the dataset reveals a leading spiral consistent with Cepheids [9], indicating that the global model may be too rigid for local orientations.

The warp's presence across both old and young populations [10] points to a universal gravitational origin, such as satellite tides or halo misalignment, and rules out hydrodynamical forces that primarily affect gas. Additionally, the high precession rate indicates a rapidly evolving structure; following Poggio et al. (2020) [7], this suggests that the exciting gravitational perturbation occurred relatively recently.

Next steps for this project are to investigate an $m=2$ model with higher-order components to map the disc's complex kinematic asymmetries more precisely (as seen in [3,8]).

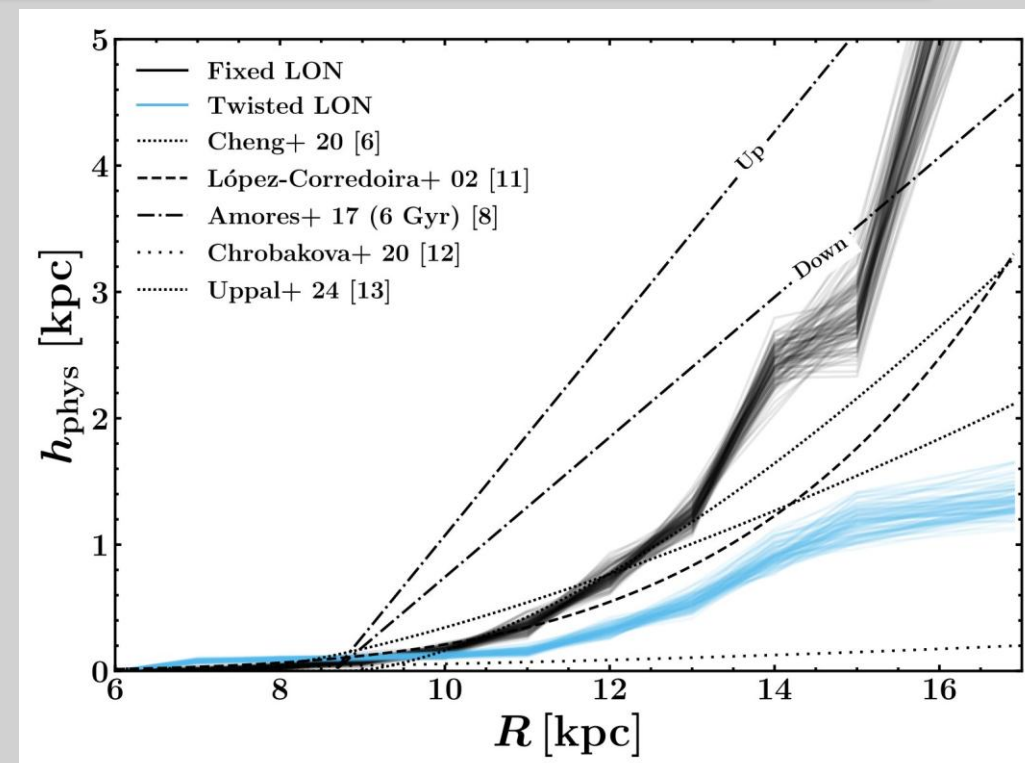


Figure 6

Data (2)

The dataset, adopted from Khanna et al. [2], comprises ~ 8.7 million Red Clump stars. These low-mass stars have evolved off the red giant branch into their core-helium-burning stage, representing the intermediate-to-old stellar population and serving as precise standard candles. Unlike younger Cepheid variables tracing the young gas disc [3], Red Clump stars trace the older stellar disc showing signs of past perturbations.

Figure 1 reveals the physical bending of the Milky Way, where the median vertical height (red line) traces a growing outer warp. Note that obscuring midplane dust blocks the view to some stars, creating low-density gaps near $Z=0$ that artificially skew this median. Since this dust obscures the true geometric shape, we must rely on stellar velocities to analyse the warp without bias.

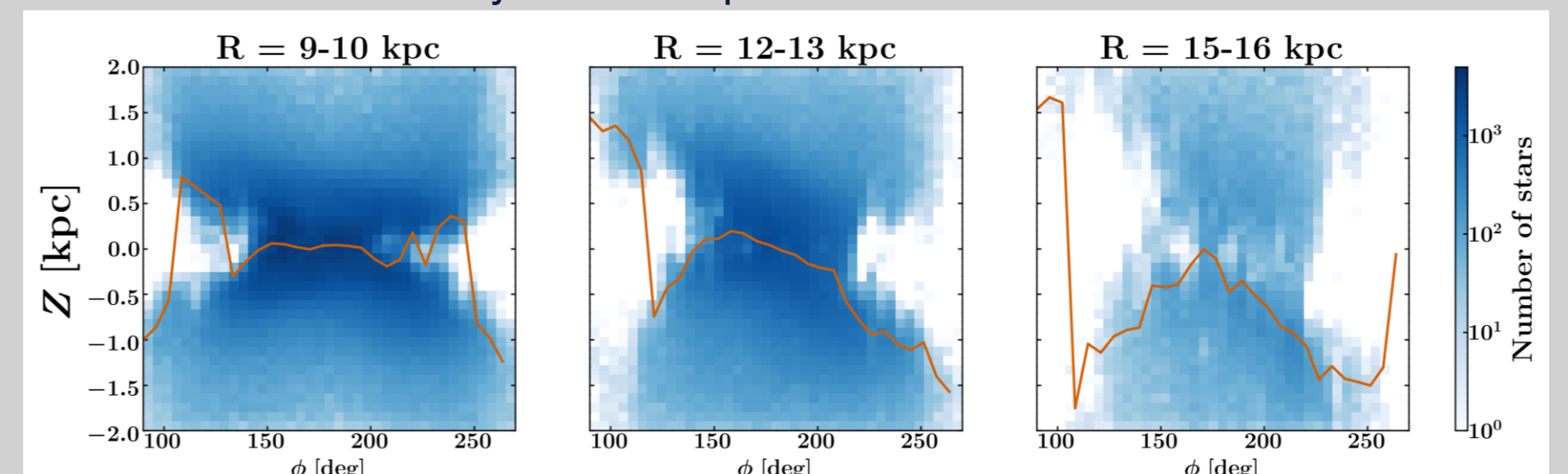


Figure 1

While the dataset contained full 3D velocity and positions for many stars, radial velocities were unavailable for ~ 4.2 million stars. To incorporate these stars into the modelling, the velocities were estimated using the assumption of zero Galactocentric radial velocity ($v_{R,gal} \approx 0$), valid for our rotation-dominated disc [4].

The resulting vertical velocity field can be seen in Figure 2. It shows distinct, coherent regions of upward and downward vertical motion, providing a direct kinematic signature of the Galactic warp.

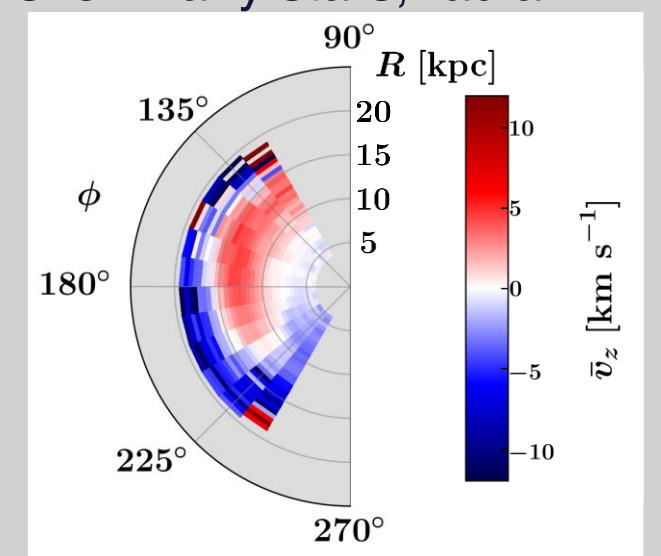


Figure 2

Results (4)

MCMC analysis was performed to fit our kinematic models to the data. Both models successfully reproduce the observed vertical velocity trends, as shown in Figure 3.

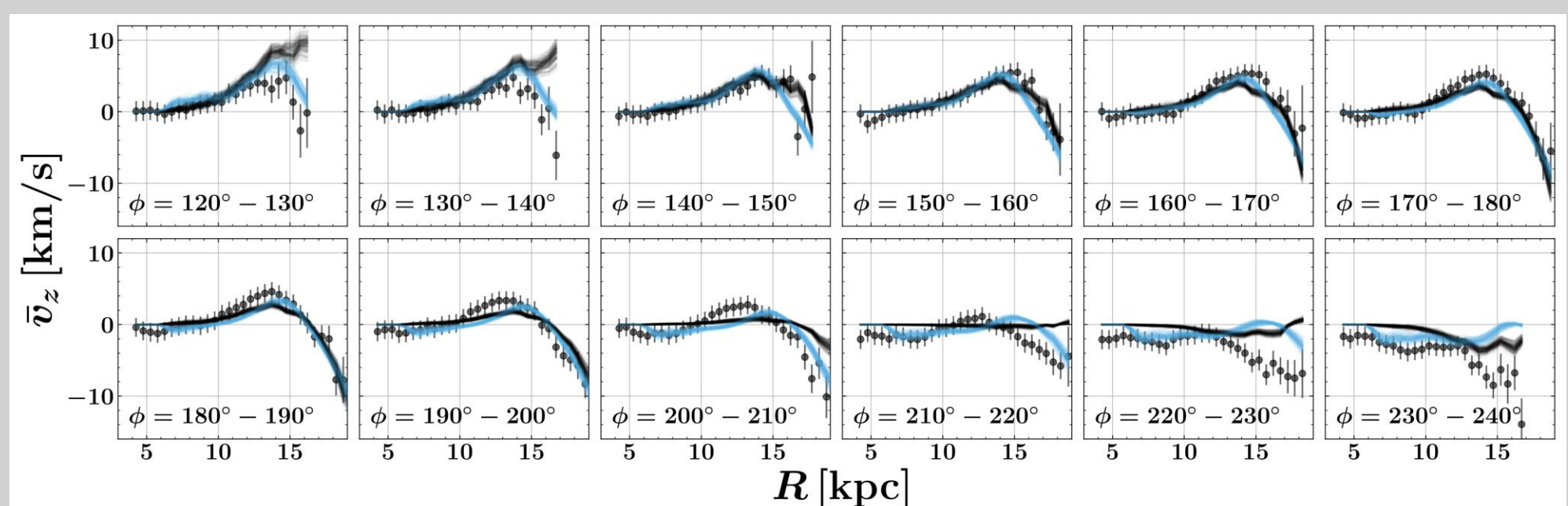


Figure 3: Black line – Fixed LON; Blue line – Twisted LON; black dots – Gaia data including estimated velocities.

Precession: A consistent prograde precession rate of $\omega_p \approx 12 \text{ km s}^{-1} \text{ kpc}^{-1}$ was found, indicating the warp completes a full rotation roughly every 0.5 Gyr.

Geometry: The fixed model places the line of nodes at $\phi'_w \approx 120^\circ$ (Figure 5, black). The twisted model (Figure 5, blue), reveals a line of nodes that varies with radius, forming a trailing spiral structure. With this, the models diverge significantly in their predicted physical height (h). The fixed model predicts a steep warp over 5 kpc, with the twisted model reaching a maximum of $h \approx 1.5 \text{ kpc}$ (Figure 4).

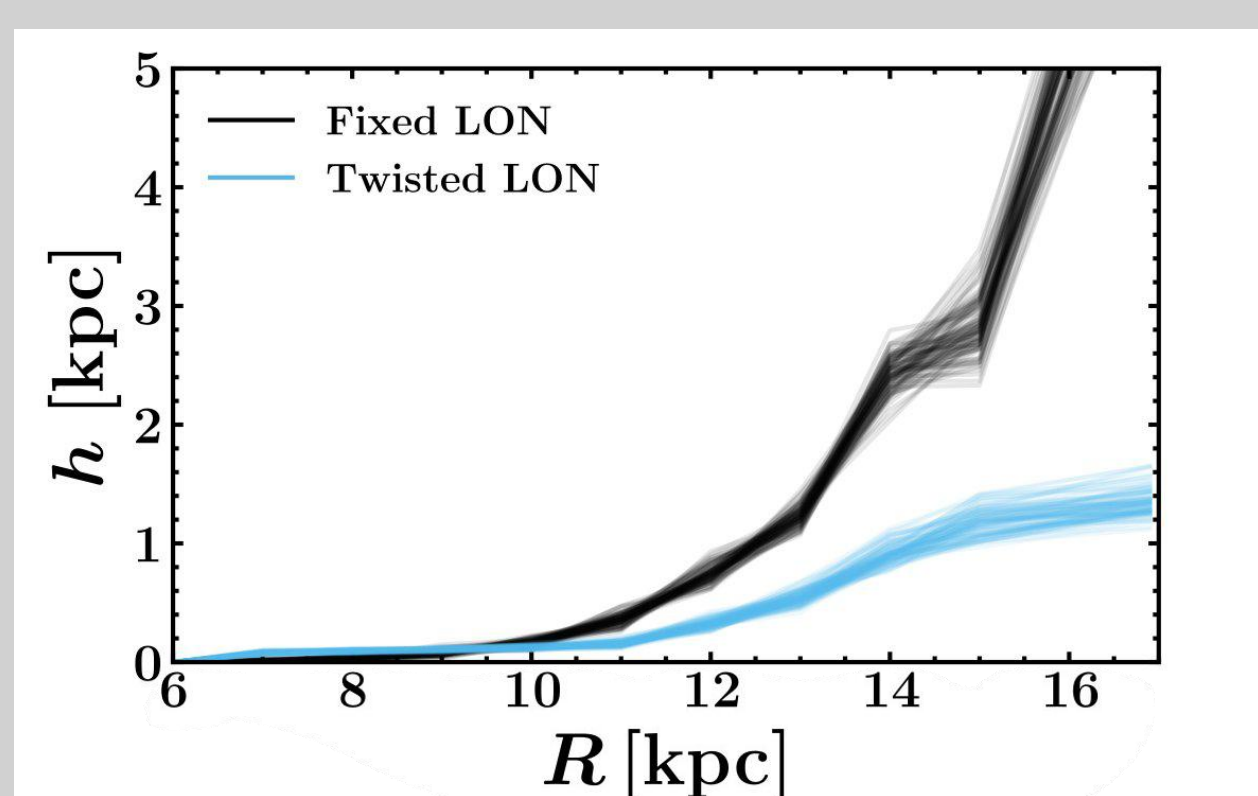


Figure 4

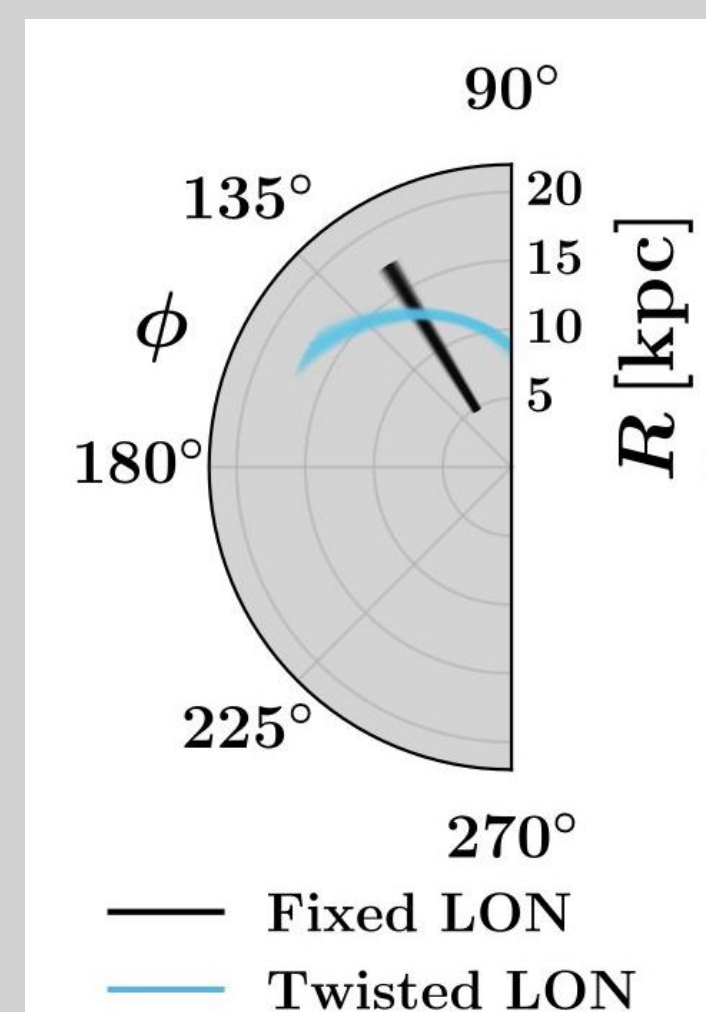


Figure 5

1. The line of nodes represents the intersection between the warped galactic disc and the flat midplane.
2. A walker is an individual agent that explores the parameter space, using the positions of its peers to efficiently find the best fit for the data.

References:

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